DAY 04 • LEARNING JOURNAL

☁

Data Analytics & BigQuery for Data Analysts

Day 4 Notes — Why Google Cloud, How BigQuery Works, and Turning Raw Data into Real Insights

Google Cloud Basics | Data Roles | Scalability | BigQuery Fundamentals | Data → Insights

Compiled from personal study notes • Google Skills: "Introduction to Data Analytics on Google Cloud"

GETTING STARTED

Data Analytics on Google Cloud

Before diving into tools, it helps to know who works with data, and what usually slows them down.

Data-Specific Roles

Three roles show up again and again in any data team. They overlap, but each one has a distinct focus — think of them as three different jobs on the same construction site.

A

Data Analyst

Derives insights from data by writing queries and building visualizations. Turns raw numbers into charts and answers that a business can act on.

SQL · BI tools · R · Python

E

Data Engineer

Designs, builds, and maintains the data pipelines that move and store data reliably, so analysts always have clean data to work with.

Background: Computer engineering

S

Data Scientist

Analyzes data and builds statistical or ML models to explain patterns and predict what will happen next, not just what already happened.

SQL · R · Python · TensorFlow

💡 **In simple words:** Data Engineers build the pipes, Data Analysts read the water meter and explain what it means, and Data Scientists predict how much water will be needed next month.

Two Common Barriers

Too much data

Query volumes grow so large that everyday tools slow down or simply cannot handle the scale, making analysis painfully slow.

Data that isn't connected

Useful information is scattered across many separate systems, so analysts spend more time hunting for data than analyzing it.

3 Three Themes Behind Most Struggles

1 · Querying

Queries take too long to run, or the logic behind them becomes very complicated.

2 · Infrastructure

Issues with scalability, ongoing maintenance, and keeping systems running smoothly.

3 · Storage

Capacity limits, rising costs, or data that isn't centralized in one place.

🔗 Data Engineers and Data Scientists run into very similar issues. This is exactly the gap that Google Cloud is designed to close — by handling storage and infrastructure, it lets every one of these roles focus on their actual job.

THE CASE FOR THE CLOUD

Why Google Cloud for Data Analysis?

Google faced these exact same data problems internally, years before most companies did — and built solutions for itself first.

> "We realized this ourselves at Google several years ago, and so we started developing systems that scale with your data growth, are managed so you aren't wasting time on complexity, and are generally awesome — so you can get back to data insights, not data management."

3 Three Reasons It Works So Well

1

Cheap, unlimited storage

Storage is inexpensive and virtually unlimited — you rarely have to worry about running out of room.

2

Focus on queries

You spend your time on questions and queries, not on managing servers or infrastructure.

3

Massive scalability

The platform expands to match sudden or long-term growth in your data, automatically.

$ The Cost of Storage Has Dropped

Year	Storage size	Approximate cost
1981	10 MB hard drive	$30,000
2023	1 TB hard drive — 100,000× more space	Less than $20

This is one hundred thousand times more storage for a fraction of the price — the economic shift that makes analyzing huge amounts of data realistic for any organization, not just tech giants.

⚙ Traditional Platforms Demand Heavy Investment

Compute | **Storage** | **Network** | **Maintenance**

What teams must manage, on-prem

- Monitoring system health and query activity
- Resource provisioning for upcoming workloads, and handling growing scale as data increases
- Reliability of the overall system
- Deployment & configuration, utilization improvements, performance tuning

⏱ Every one of these jobs takes time away from actual analysis. Google Cloud's promise is simple: let Google manage this wheel, so your team can focus on insights.

CORE CONCEPT

Google Cloud Enables On-Demand Scalability

The single biggest shift moving from your own servers to the cloud: you stop guessing how much capacity you'll need.

When you own your hardware, you must decide in advance how many machines to buy. Get it wrong, and you either waste money or wait far too long for results.

ON-PREMISES

Overprovisioned

1 second using 120 idle machines — demand < capacity. You paid for 120, but most sit unused most of the time.

Underprovisioned

120 seconds using just 1 machine — demand > capacity. Everyone waits, because there isn't enough power to go around.

GOOGLE CLOUD

Elastic, matched to demand

As demand rises, the cloud simply adds more virtual machines to match it — automatically, in real time.

≈1 second using 120 machines working in parallel — the same query, matched to demand instead of a fixed budget.

💡 **In simple words:** Instead of guessing whether you need 1 machine or 120, Google Cloud gives you a pool of virtual machines that automatically grows or shrinks to match exactly what your query needs, right now. A query that would take 120 seconds on one machine can finish in about 1 second using 120 machines together.

Scaling Is Fully Automatic

There is no guesswork for demand — if you need more power, the cloud scales up elastically on its own, and scales back down once the work is done, so you're never paying for idle capacity for long.

Decoupling Storage and Compute

One key feature of Google Cloud: storage and compute are separated instead of living on the same machine. On-premises, both are usually co-located on the same server — so you pay for full processing power even when no queries are running.

ON-PREMISES

Pay for the ability to use processing power, even when no queries are actively running.

GOOGLE CLOUD

Pay only for the resources you're actually using — storage and compute are billed and scaled independently.

MEET THE PRODUCT

From Data to Insights with BigQuery

Google Cloud's enterprise data warehouse, purpose-built for analytics.

◆ **What is BigQuery?** BigQuery is Google Cloud's fully managed, petabyte-scale enterprise data warehouse that helps you manage and analyze your data using familiar SQL — with zero infrastructure to set up or maintain.

📚 **Analogy:** If your organization's data were a library, an on-premises warehouse is one you personally built, staffed, and maintain — you handle everything from shelving to security. BigQuery is a fully staffed national library: you simply ask your question at the desk (write a query), and the answer comes back, no matter how many books (rows of data) are involved.

" Why It's Called an "Enterprise Data Warehouse"

↕

Gigabyte to petabyte scale

Handles storage and SQL queries at any scale — from a spreadsheet's worth of data to petabytes.

✦

Built-in ML & GIS

Run machine-learning models and geographic analysis directly with SQL, no separate tools needed.

🔒

Encrypted & available

Your data is encrypted, durable, and always accessible when you need it.

⚡

Serverless & real-time

No servers to manage, plus the ability to run real-time analytics on streaming data.

✓ Fully-Managed, By Design

No-Ops

You never manage servers, patches, or backups yourself.

Serverless

Google runs and scales the infrastructure behind the scenes.

Petabyte-scale

Designed from the ground up to handle enormous datasets.

BigQuery's serverless architecture lets you use familiar SQL queries to answer your organization's biggest questions — with zero infrastructure management. You can query terabytes of data in seconds, and petabytes in minutes.

🔑 **Key idea — maximum flexibility:** BigQuery maximizes flexibility by separating the compute engine that analyzes your data from your storage choices. This is the same "decoupling" concept from the previous page, applied directly inside the product.

KEY CHARACTERISTICS

Eight Reasons BigQuery Stands Out

Together, these traits explain why analysts trust BigQuery for serious, everyday work.

1

Reliable

Backed by Google's own global data centers — the same infrastructure that runs Search and Gmail.

2

Economical

Pay only for what your query processes, or choose flat-rate capacity pricing for predictable costs.

3

Secure

Data access controlled through Access Control Lists (ACLs) and Identity and Access Management (IAM).

4

Auditing

Cloud Audit Logs track admin activity, data access, and system events for compliance.

5

Scalable

Virtually unlimited storage and compute, using a parallel, distributed model for fast queries.

6

Flexible

Mash up data across multiple datasets, and stream real-time data in as it happens.

7

Easy to use

Built on familiar SQL, with no indexes to manage, and support for open standards.

8

Public datasets

Ready access to a large library of public datasets to practice and explore with.

💡 **In simple words:** Reliable, economical, secure, auditable, and scalable are the qualities every enterprise tool needs. Flexible, easy to use, and public datasets are what make BigQuery genuinely pleasant to work in day-to-day — even for someone new to data analysis.

HOW YOU ACCESS IT

Interfacing with BigQuery

There's more than one way to talk to BigQuery, depending on how you like to work.

Web UI

A visual browser interface in Google Cloud Console — write and run SQL, see results instantly, no setup needed.

Command-line (CLI)

The "bq" command-line tool runs queries and manages resources from a terminal — great for automation.

REST API

Lets developers integrate BigQuery directly into their own applications and custom software.

GUI SQL editors

Third-party SQL editing tools that connect to BigQuery for a familiar, feature-rich experience.

2 BigQuery Is Really Two Services in One

This split is exactly the storage/compute decoupling explained earlier — it's what allows each side to be optimized and scaled independently.

Conceptual diagram — Google's high-speed internal network connects the two services.

BigQuery Managed Storage

Stores your structured data safely and cost-effectively, compressed and organized for fast reads — no disks to manage.

PETABIT NETWORK

BigQuery Analytics Engine

The distributed compute layer that runs your SQL queries across many machines in parallel, at any scale.

Querying data from different sources

BigQuery supports querying data from a variety of sources — not just data already loaded into it. This includes data sitting in Cloud Storage, spreadsheets, and other external systems, so you don't always need to move data in first before you can analyze it.

THE BIG PICTURE

From Raw Data to Real Insights

Every data analyst task fits into one of five stages — and BigQuery offers a scalable tool for each one.

① Ingest
Get data in from a variety of formats and sources.
Risk: volume, variety, velocity

→

② Transform
Prepare, clean, and transform data quickly.
Risk: slow processing, unclear logic

→

③ Store
Create, save, and store datasets inexpensively.
Risk: cost, scale, latency

→

④ Analyze
Derive insights at scale, no servers to manage.
Risk: slow queries, high volume

→

⑤ Visualize
Explore and present impactful insights.
Risk: size, latency, stale data

💡 **Why this matters:** Each of these five tasks has its own challenges — analysis may get harder simply because queries run slowly on huge volumes of data, or because business logic is genuinely complicated. Together, they make it harder for analysts to get useful, timely, actionable insight — precisely what BigQuery's toolset is designed to solve, at every stage.

Course Agenda

1. **BigQuery for Data Analysts**
   Course overview
2. **Exploring & preparing your data with BigQuery**
3. **Cleaning & transforming your data**
4. **Ingesting & storing new BigQuery datasets**
5. **Visualizing your insights from BigQuery**
6. **Developing scalable data transformation pipelines with Dataform**
7. **BigQuery Studio**

⚾ **Real-world use case — Major League Baseball (MLB):** MLB is a real-world example of a company transformed through analytics on the cloud — using BigQuery to analyze huge volumes of game and player data at scale, turning raw statistics into insights that shape strategy, fan engagement, and broadcasting decisions.

WRAPPING UP

Key Takeaways & Glossary

A quick recap of everything covered today, plus simple definitions for the terms used most.

✓ Today's Key Takeaways

- Data Analysts, Data Engineers, and Data Scientists each play a distinct role, but all three face similar barriers around querying, infrastructure, and storage.
- Google Cloud removes the need to guess capacity by offering cheap, virtually unlimited storage and on-demand, elastic scalability.
- Decoupling storage from compute means you only pay for what you actually use — a core idea behind both Google Cloud and BigQuery.
- BigQuery is a fully-managed, serverless, petabyte-scale enterprise data warehouse built for fast SQL analytics.
- BigQuery is reliable, economical, secure, auditable, scalable, flexible, and easy to use — with ready-to-use public datasets.
- You can interface with BigQuery through the Web UI, CLI, REST API, or third-party GUI SQL editors.
- BigQuery is really two services working together: managed storage and a powerful analytics engine.
- Every analyst task — ingest, transform, store, analyze, visualize — maps to a scalable BigQuery tool built to overcome its unique challenges.

A-Z Glossary — Plain-English Definitions

Term	Definition
Serverless	You never see or manage the underlying servers — the cloud provider handles all of it invisibly.
Compute	The processing power used to run calculations, such as executing a SQL query.
Elastic scaling	Resources automatically grow when demand rises, and shrink back down when demand falls.
Petabyte	A unit of data equal to about 1 million gigabytes — enough to store roughly 500 billion pages of text.
IAM	Identity and Access Management — controls exactly who can see or use which resources.
ACL	Access Control List — a rulebook that defines specific permissions on a specific resource.
Data warehouse	A central system built to store and analyze large amounts of data from across an organization.
Public dataset	A ready-made dataset made freely available by BigQuery for anyone to practice and explore with.

→ **Looking ahead:** Day 5 builds directly on this foundation — moving from understanding what BigQuery is, into actually exploring, cleaning, and transforming real data inside it.